# Module 1: Basic Concepts — Practice

## Lesson 2 | Apply the R fundamentals

### Exercise 1: Greeting Script in R

**Question A:** Construct an R script that outputs the greeting “Hello, World!” Then, alter the script to replace “World” with your own name.

**Solution A:**

**Question B:** Utilize the `cat()` function to display “Hello, World!” in the console. Observe and report any variations in the output compared to the `print()` function.

**Solution B:**

### Exercise 2: Storing Personal Favorites

Store your preferred number, color, and a boolean value that signifies your liking for pizza. Display these stored values.

**Solution:**

### Exercise 3: Types and Collections

**Question A:** Generate a vector with the numbers from 1 to 10. Determine and document the vector’s type as well as the types of the elements within it.

**Solution A:**

**Question B:** Form a list holding a numeric value, a character string, and a boolean value. Ascertain the list’s type and the types of its individual elements.

**Solution B:**

### Exercise 4: Checking Number Sign

Compose a script that discerns whether a given number (hard-coded in the script) is positive, negative, or zero, and relays an appropriate message.

**Solution:**

### Exercise 5: Iterative Loops

**Question A:** Write a `for` loop that computes and exhibits the squares of numbers from 1 to 10.

**Solution A:**

**Question B:** Implement a `while` loop to effectuate a countdown from 10 to 1, displaying each countdown step.

**Solution B:**

### Exercise 6: Area Calculation Function

Craft a function named `calculate_area` that accepts a circle's radius and returns its area. Check the function's accuracy with a variety of radii.

**Solution:**

### Exercise 7: Vector Mathematics

**Question A:** Form two vectors, a and b, each with 5 random numbers. Execute and print the results of element-wise addition, subtraction, multiplication, and division.

**Solution A:**

**Question B:** Use comparison operators to compare the elements of a and b and communicate the outcomes.

**Solution B:**

### Exercise 8: Working with Factors

Institute a factor variable that delineates three kinds of fruit. Illustrate the factor variable, its levels, and transmute it into numerical values.

**Solution:**

### Exercise 9: Data Frame Adjustments

Forge a modest data frame that enlists student IDs, names, and scores. Accomplish these tasks:

- Select and unveil only the students' names and scores.
- Incorporate a column indicating if the student passed (score >= 50).

**Solution:**

### Exercise 10: Interactive User Query (Optional)

Write code that asks the user to enter his name and age, then returns a message along with a mention of whether they are above the age of 18.

**Solution:**

### Home Work

**Scenario:** You are a Data Scientist working for a consulting firm. One of your colleagues from the Auditing department has asked you to help them assess the financial statement of organization X.

You have been supplied with two vectors of data: monthly revenue and monthly expenses for the financial year in question. Your task is to calculate the following financial metrics:

1. **Profit for each month**: Calculate the profit for each month by subtracting expenses from revenue.
2. **Profit after tax for each month**: The tax rate is 30%. Calculate the profit after tax for each month.
3. **Profit margin for each month**: The profit margin is equal to the profit after tax divided by revenue, expressed as a percentage.
4. **Good months**: Identify the months where the profit after tax was greater than the mean profit after tax for the year.
5. **Bad months**: Identify the months where the profit after tax was less than the mean profit after tax for the year.
6. **The best month**: Identify the month where the profit after tax was maximum for the year.
7. **The worst month**: Identify the month where the profit after tax was minimum for the year.

**Additional Requirements:**

- All results need to be presented as vectors.
- Results for dollar values need to be calculated with $0.01 precision but presented in units of $1,000 (i.e., 1k) with no decimal points.
- Results for the profit margin ratio need to be presented in units of percentage with no decimal points.
- Note: It is okay for tax for any given month to be negative (in accounting terms, negative tax translates into a deferred tax asset).

```
# c() combines compatible values into a vector in the supplied order.
# Assignments save the vector under a name; printing that name displays its current
 ↪  values.
# Given data
revenue <- c(14574.49, 7606.46, 8611.41, 9175.41, 8058.65, 8105.44, 11496.28,
 ↪  9766.09, 10305.32, 14379.96, 10713.97, 15433.50)
expenses <- c(12051.82, 5695.07, 12319.20, 12089.72, 8658.57, 840.20, 3285.73,
 ↪  5821.12, 6976.93, 16618.61, 10054.37, 3803.96)
```